SYNTOMICITY IS PRESERVED BY A FLAT SURJECTIVE MORPHISM OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's informal phrasing 'Thus may use Avramov's result' is quoted verbatim.

Editorial note. The counted spans carry navigation stubs for the references leaving them; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Permanence of finite presentation along a flat surjective morphism of finite presentation, and of flatness. Same chapter, Lemma lemma-flat-finitely-presented-permanence and Morphisms, Lemma morphisms-lemma-flat-permanence, invoked at proof lines 4731-4734. Gives that q is of finite presentation and flat.
- Characterisation of syntomic morphisms by local complete intersection fibres. Morphisms, Lemma morphisms-lemma-syntomic-locally-standard-syntomic, invoked at proof lines 4735-4738. Reduces syntomicity of q and f to a statement about the local rings of the fibres.
- Avramov's theorem on flat local homomorphisms whose target is a complete intersection. Divided Power Algebra, Lemma dpa-lemma-avramov, invoked at proof lines 4747-4751. Concludes that both O_{Y,y} and the fibre ring O_{X,x}/m_y O_{X,x} are complete intersections.

Source reference. Original lemma 09Q7: sources/stacks-chow/tags.txt, source lines 12275–12275. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

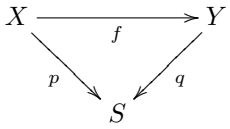
Source reference. Original lemma 02KL: sources/stacks-descent/pinned-original-source.tex, source lines 4602–4602. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02JZ: sources/stacks-chow/tags.txt, source lines 3354–3354. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01UE: sources/stacks-chow/tags.txt, source lines 2407–2407. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let*



be a commutative diagram of morphisms of schemes. Assume that

(1) *f is surjective, flat, and locally of finite presentation,*
(2) *p is syntomic.*

Then both q and f are syntomic.

Proof. By Lemma 02KL we see that q is of finite presentation. By Morphisms, Lemma 02JZ we see that q is flat. By Morphisms, Lemma 01UE it now suffices to show that the local rings of the fibres of $Y \to S$ and the fibres of $X \to Y$ are local complete intersection rings. To do this we may take the fibre of $X \to Y \to S$ at a point $s \in S$, i.e., we may assume S is the spectrum of a field. Pick a point $x \in X$ with image $y \in Y$ and consider the ring map

$$\mathcal{O}_{Y,y} \longrightarrow \mathcal{O}_{X,x}$$

This is a flat local homomorphism of local Noetherian rings. The local ring $\mathcal{O}_{X,x}$ is a complete intersection. Thus may use Avramov's result, see Divided Power Algebra, Lemma 09Q7, to conclude that both $\mathcal{O}_{Y,y}$ and $\mathcal{O}_{X,x}/\mathfrak{m}_y\mathcal{O}_{X,x}$ are complete intersection rings. □